Abd Alrahman Ahmed Mousa Mohamed

Computer Science · Software Developer

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> Software developer focused on desktop applications and data analysis pipelines. Work is delivered with runnable demos, screenshots, and client-ready reports rather than slides alone.

Projects

1. Cosmic Observer

Windows astronomy desktop app — 3D sky, NASA data, AI tutor.

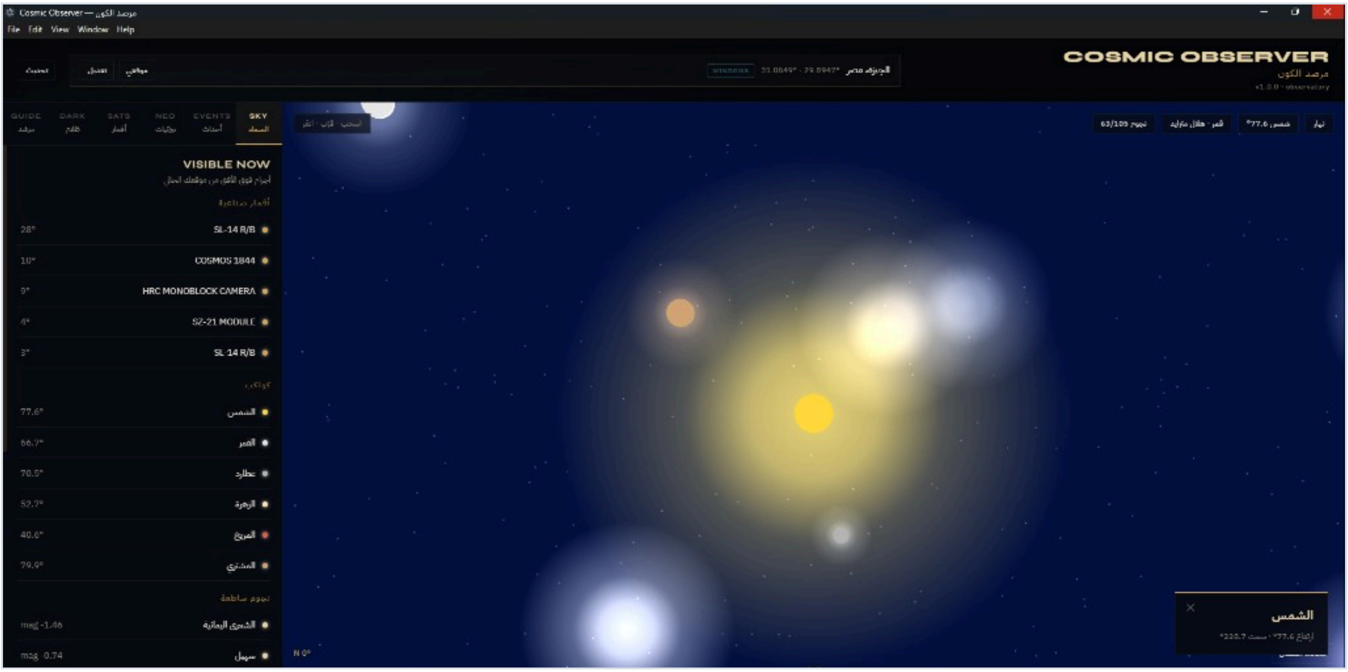
2. Vertical Farm Analysis

Python pipeline — farm telemetry to charts and PDF report.

PROJECT 01

Cosmic Observer

Windows desktop astronomy application with an interactive 3D sky, near-Earth object tracking, orbital debris monitoring, a dark-matter research module, and an AI astronomy tutor. Shipped as a native Windows installer and portable executable.



Capabilities

- Live 3D sky from GPS or manual location
- Satellites, planets, and bright stars list
- Sun / Moon status and sky events
- Modules: Sky, Events, NEO, Sats, Dark, Guide
- Arabic and English interface

Links

- github.com/engboda6/cosmic-observer
- License: MIT
- Platform: Windows (.exe)
- Demo video available on portfolio site

Cosmic Observer — Stack & Features

Built with JavaScript throughout: React and Three.js for the interface and sky rendering, Node.js/Express for local APIs and orbital calculations, Electron for desktop packaging.

Technologies

JavaScript · React 18 · Three.js · Vite · Node.js · Express · astronomy-engine · OpenAI SDK · Electron 33 · GLSL · PowerShell

Features

- Interactive 3D celestial sphere
- Sunrise/sunset and moon phases
- NASA NeoWs near-Earth object tracking
- CelesTrak orbital debris / TLE
- Dark-matter lensing research views
- Optional OpenAI astronomy tutor

Data sources

- NASA NeoWs
- JPL Sentry / CAD
- NASA CNEOS Fireball
- CelesTrak (TLE)
- VizieR / CDS

Structure

- `electron/` — desktop shell
- `server/` — local APIs
- `src/` — React UI + Three.js

PROJECT 02

Vertical Farm Telemetry Analysis

Pure Python analysis system for vertical-farm telemetry. Processes 600 production rows into a clean dataset, channel-level analysis, charts, Excel exports, and a three-page PDF report.

Pipeline

1. **prepare_dataset.py** — import and clean `vertical_farm_biomass_telemetry_ledger.csv`; parse Stack / Rack / Row IDs.
2. **analysis.py** — nutrient decay matrix for 60 irrigation channels; weak circulation; harvest-weight index; optimal feeding balances.
3. **charts.py** — EC distribution, circulation vs decay, heatmap, harvest index.
4. **report.py** — three-page PDF prepared by Abd Alrahman Ahmed.
5. **run_pipeline.py** — runs the full workflow in one command.

PROJECT 02 · DELIVERY

Libraries, Outputs & Contact

The project is written entirely in Python and converts a raw farm CSV into analysis tables, visualizations, and a report suitable for client delivery.

Libraries

- **pandas** — data cleaning and aggregation
- **numpy** — numerical calculations
- **matplotlib** — charts and PDF figures
- **openpyxl** — Excel export

Deliverables

- Clean structured dataset
- Analysis across 60 irrigation channels
- Charts and Excel workbooks
- Three-page nutrient analysis PDF

Contact

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